

A close-up, top-down view of a white ceramic coffee cup filled with a latte. The coffee has a thick layer of light-colored foam on top, decorated with intricate latte art in a darker brown color. The background is a soft, out-of-focus grey, suggesting a table or counter.

Brewing Insights: Coffee Shop Sales Analysis

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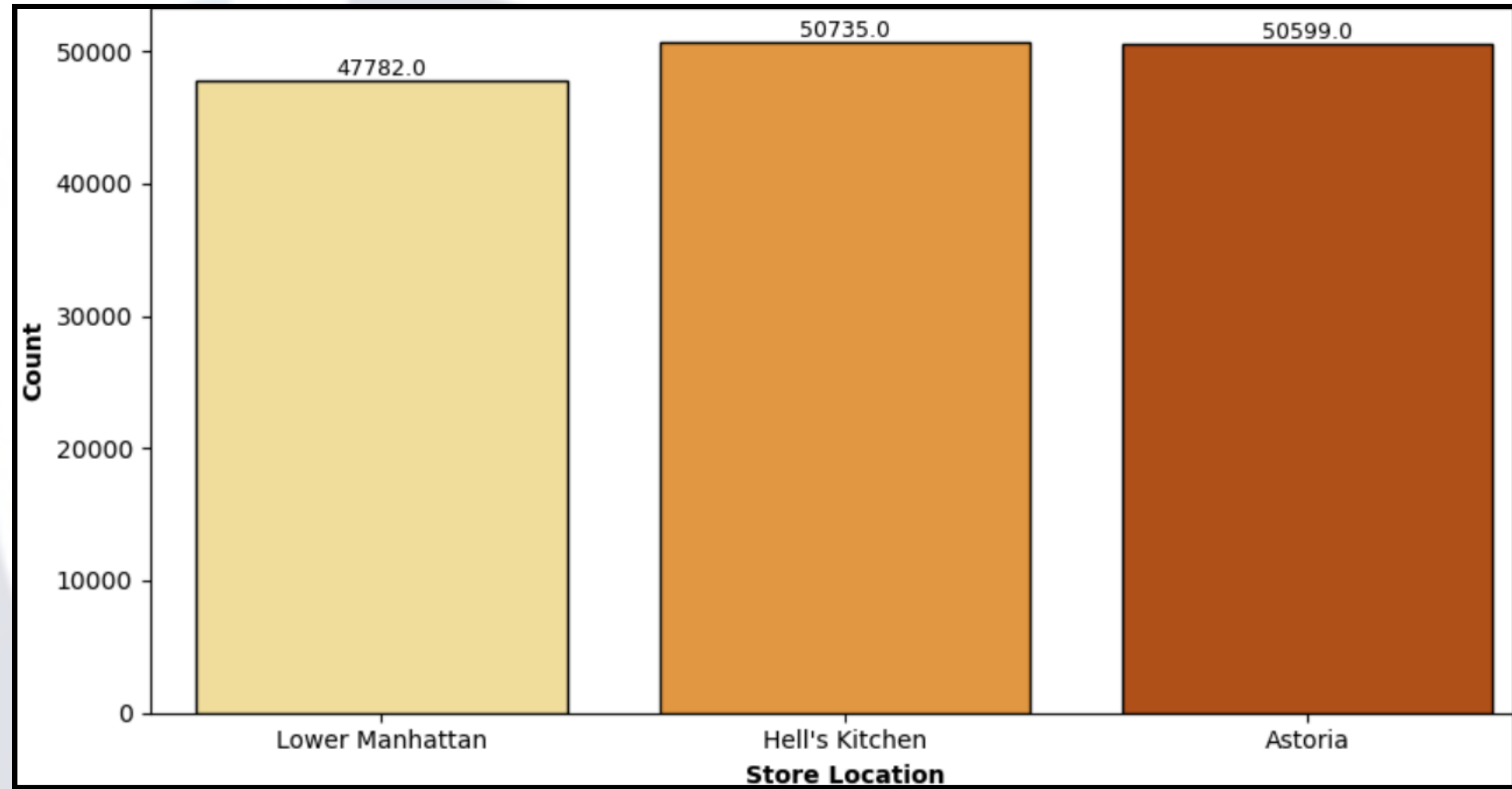
Introduction

This project focuses on analyzing coffee shop sales data to understand customer purchasing behavior, product performance, sales trends, and store-wise revenue insights.

Various data analysis and visualization techniques were used using Python libraries such as Pandas, Matplotlib, and Seaborn to generate meaningful business insights.

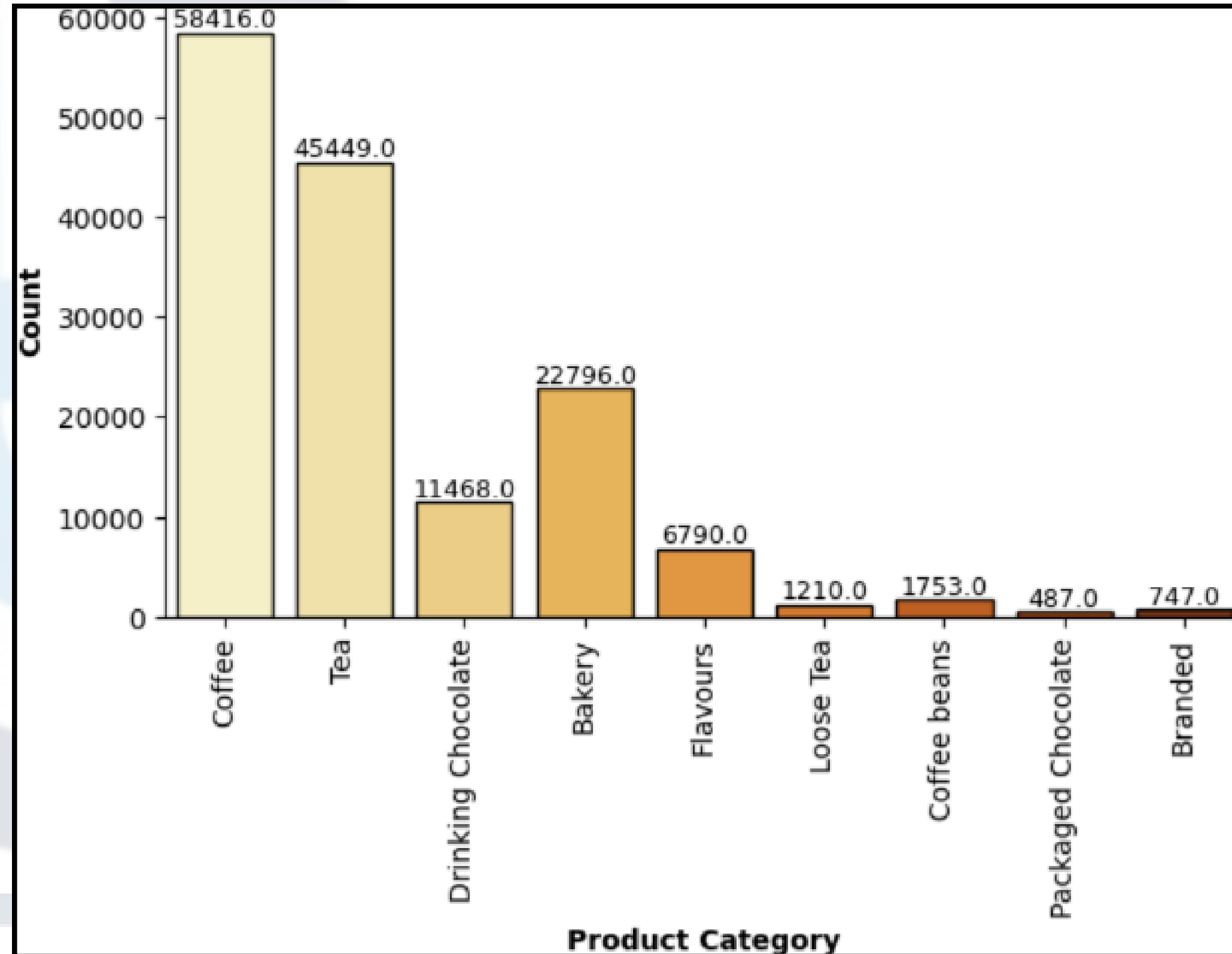
Store Location Distribution

- **Hell's Kitchen** has the highest number of transactions with around **50,735 sales**.
- **Astoria** is very close with about **50,599 transactions**, showing similar store performance.
- **Lower Manhattan** has the lowest count with around **47,782 transactions** compared to the other two locations.
- Overall, the transaction distribution across all store locations is fairly balanced, with only a small difference between stores.



Product Category Count

- **Coffee** is the most purchased category with around **58,416** transactions, showing the **highest customer demand**.
- **Tea** is the second most popular category with about **45,449** sales.
- **Bakery** products also perform well with nearly **22,796** transactions.
- Categories like **Packaged Chocolate (487)** and **Branded products (747)** have the lowest sales contribution.



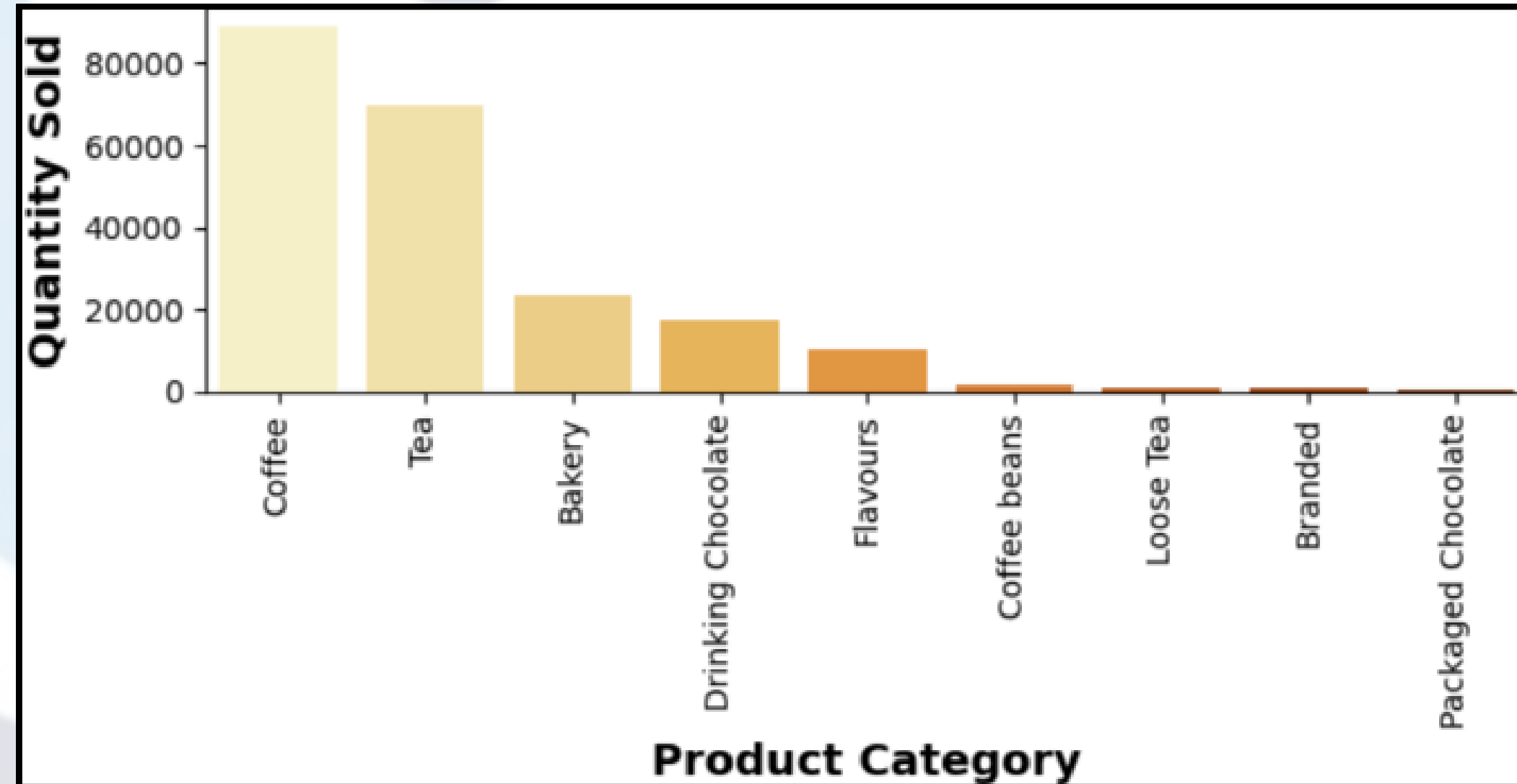
Total Sales by Store Location

- **Hell's Kitchen** generated the highest total sales, slightly **above 235,000**, making it the best-performing store location.
- **Astoria and Lower Manhattan** also show very similar sales performance, both close to **230,000+ sales**.
- The difference in sales among all three stores is very small, indicating balanced business performance across locations.



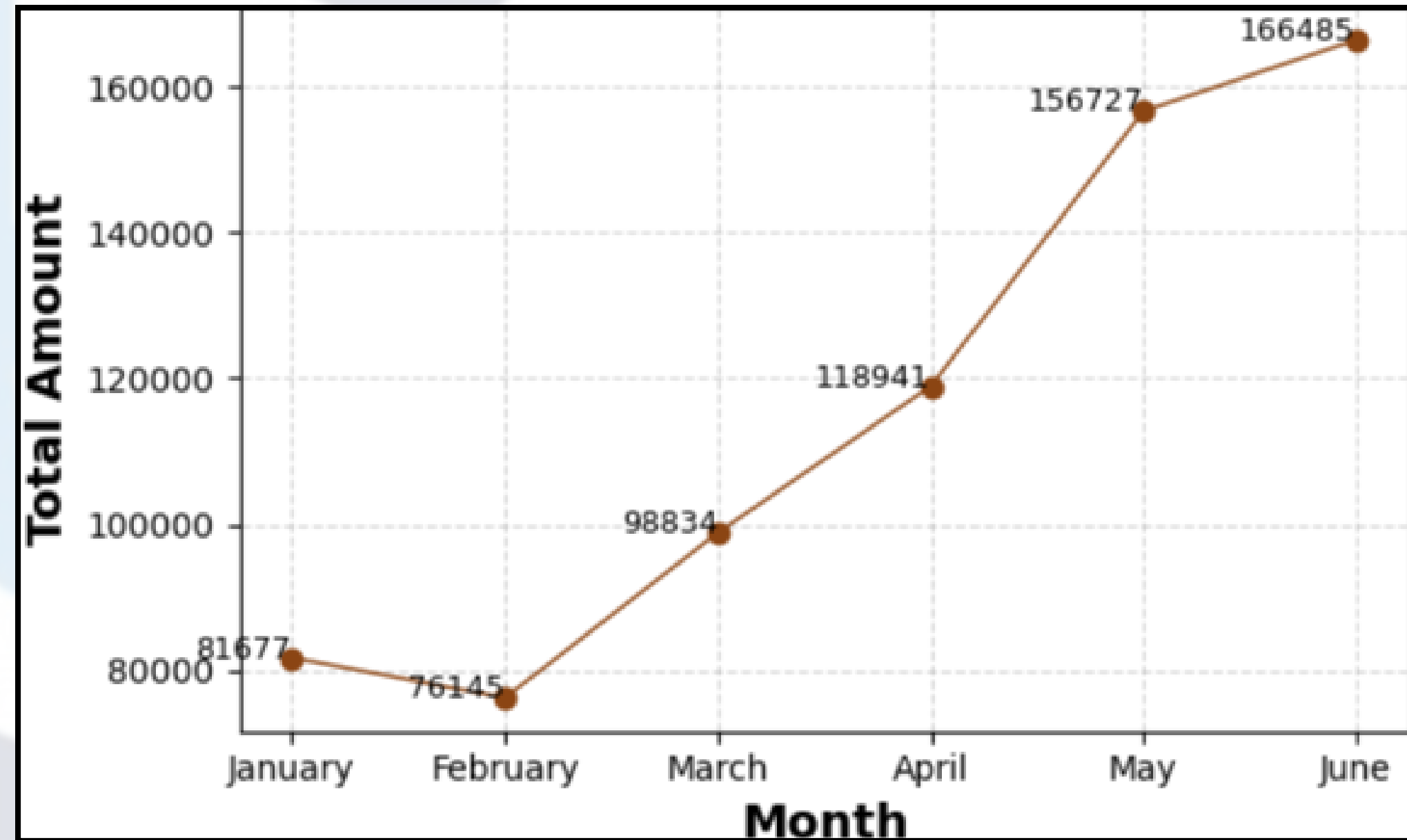
Quantity Sold by Product Category.

- **Coffee** has the highest quantity sold with nearly **90,000 units**, making it the most demanded **product category**.
- **Tea** is the second highest-selling category with around **70,000 units sold**.
- **Bakery and Drinking Chocolate** show **moderate sales** performance compared to beverages.
- Categories like **Coffee Beans, Loose Tea, Branded, and Packaged Chocolate** have very **low sales** quantities.



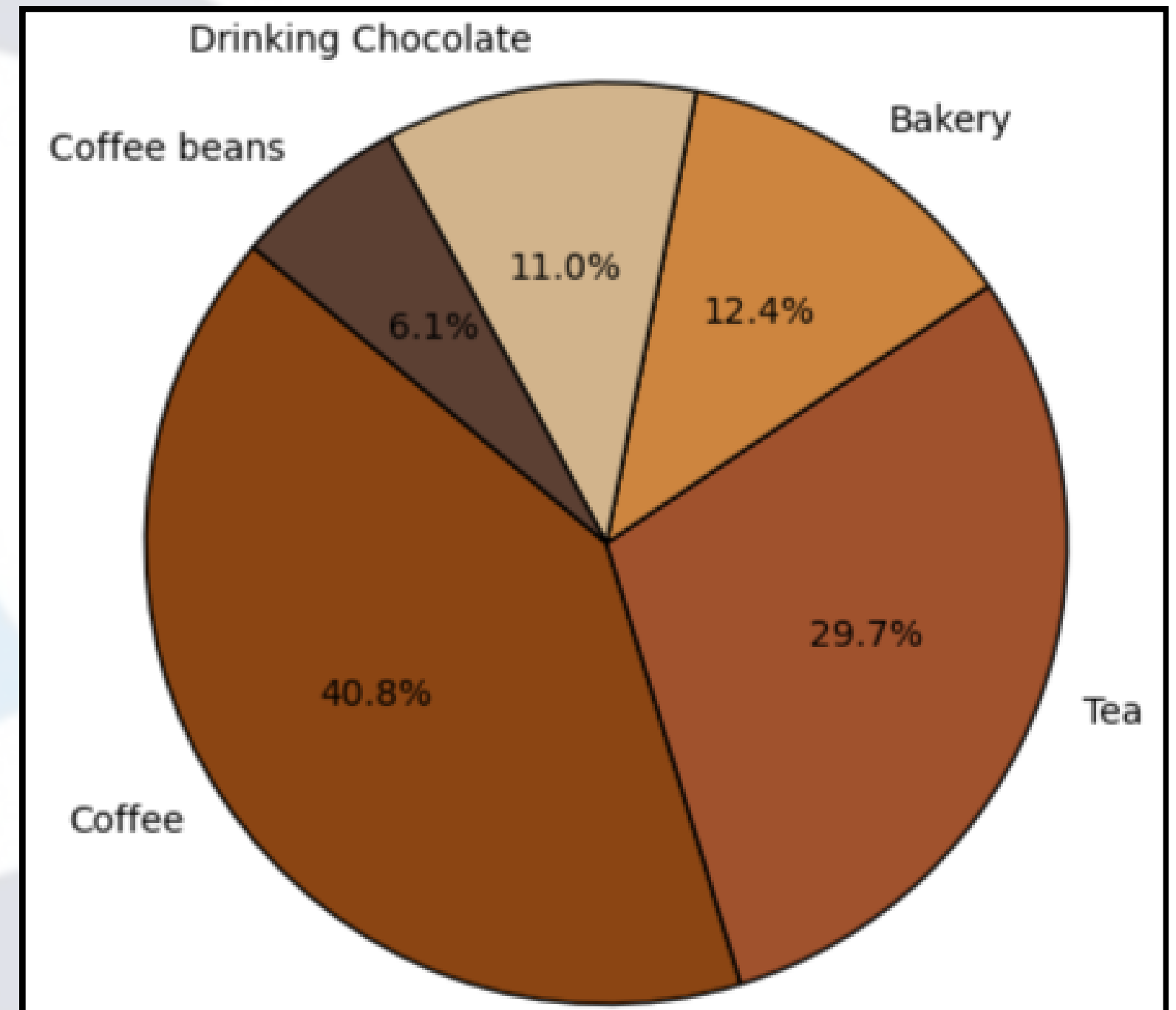
Monthly Sales Trend

- Sales slightly decreased from **81,677 in January** to **76,145 in February**.
- From March onwards, total sales increased steadily, reaching **98,834 in March** and **118,941 in April**.
- A sharp growth is observed in **May (156,727)** and **June (166,485)**, showing strong business performance in later months.



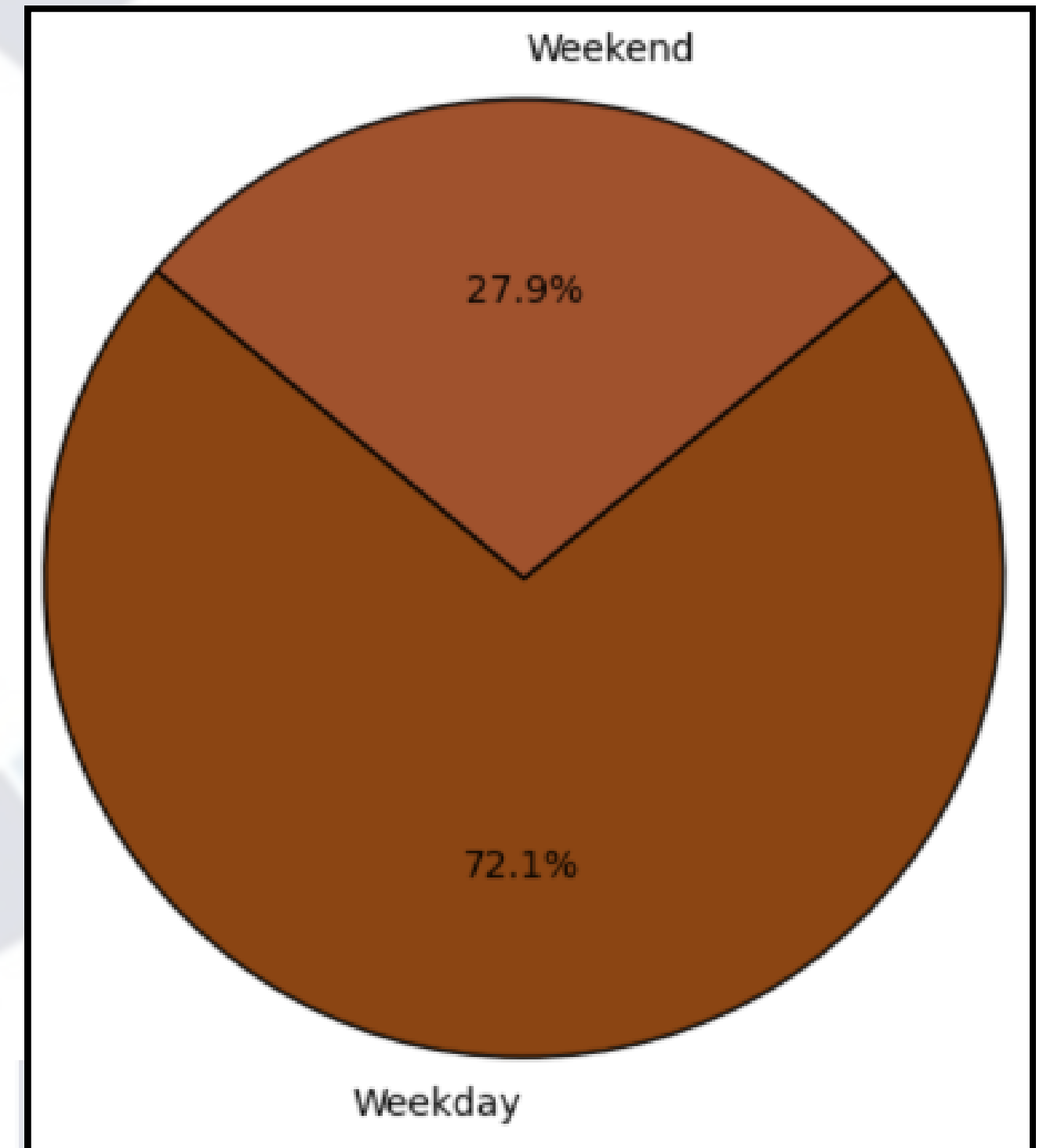
Sales Distribution by Product Category

- Coffee contributes the highest sales share with 40.8%, making it the top-selling category.
- Tea is the second highest contributor with 29.7% of total sales.
- Bakery products account for 12.4% of sales, showing moderate customer demand.
- Drinking Chocolate contributes 11.0% to overall sales.
- Coffee Beans have the lowest contribution at 6.1%, indicating lower direct purchase demand.
- Coffee and Tea together contribute around 70% of total sales, showing beverages dominate the business revenue.



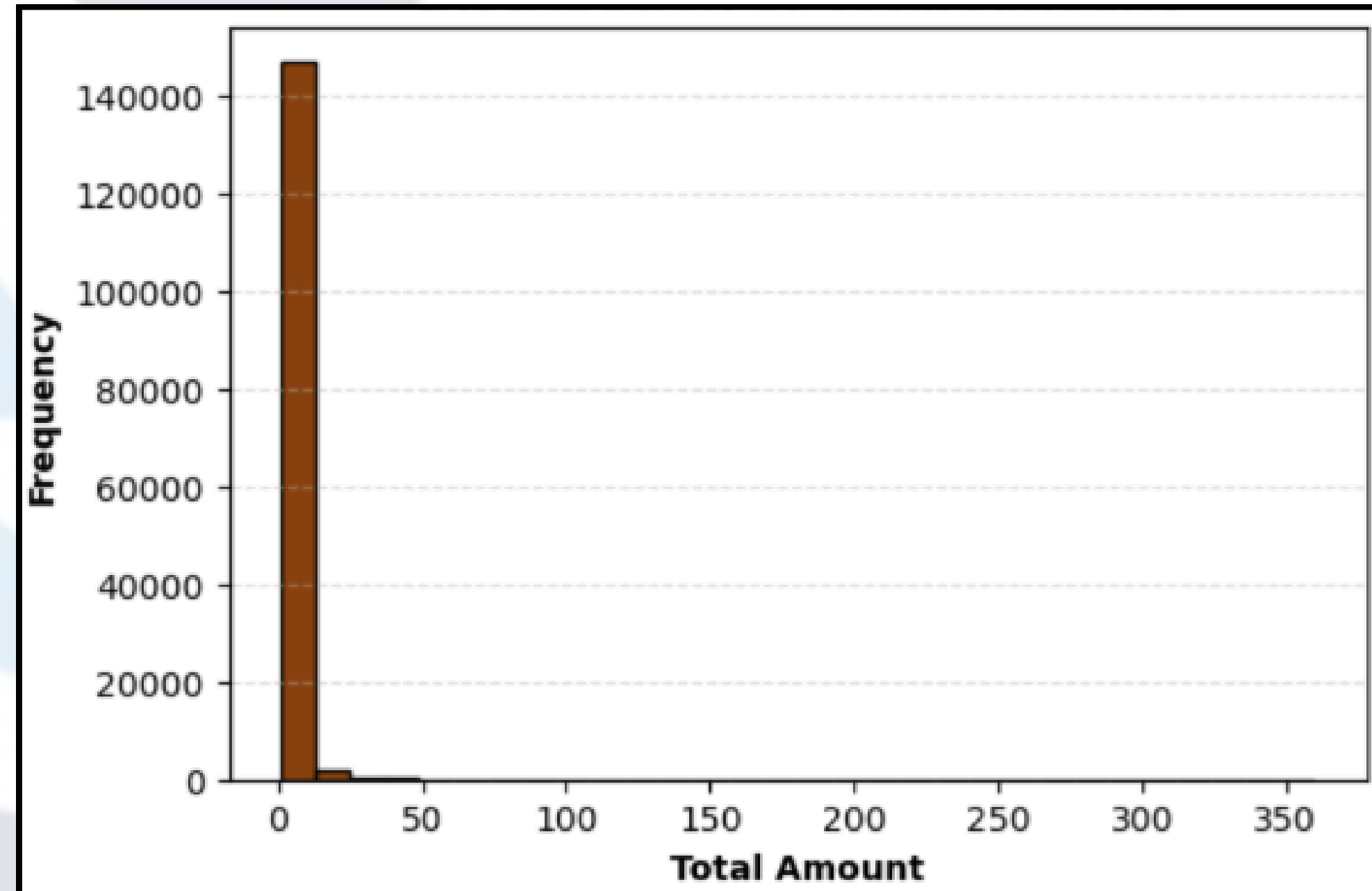
Weekday vs Weekend Sales

- Weekday sales contribute 72.1% of total revenue, showing most customer purchases happen on weekdays.
- Weekend sales account for 27.9% of total sales, which is significantly lower than weekdays.
- The coffee shop experiences higher business activity during weekdays, possibly due to office-goers and daily customers.
- Weekday sales are nearly 2.5 times higher than weekend sales.



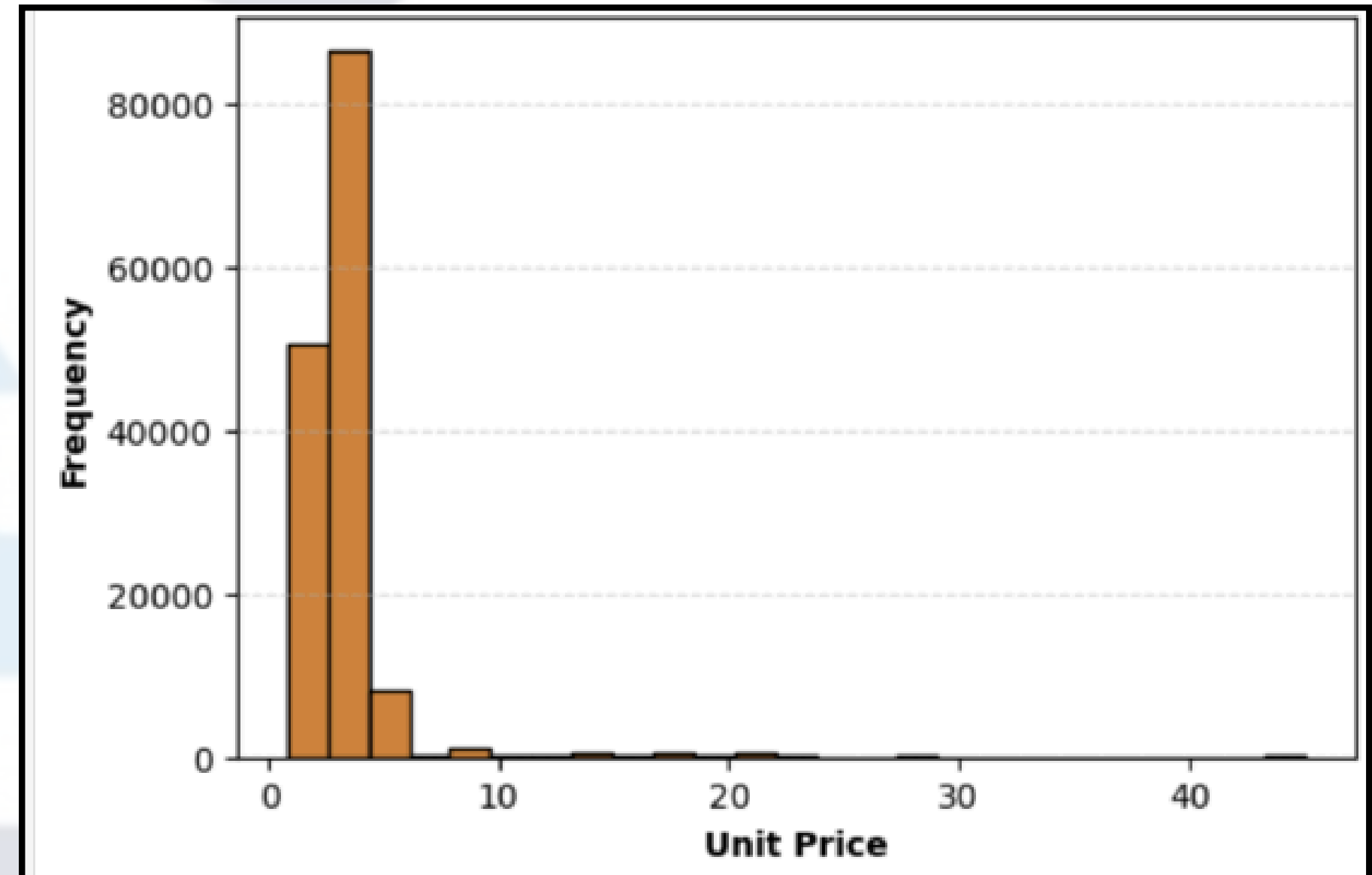
Distribution of Total Amount

- Most transactions are concentrated in the lower total amount range, indicating customers usually make small purchases.
- The histogram is highly right-skewed, showing a few transactions have very high total amounts.
- The majority of sales occur below the lower price range, while high-value purchases are rare.
- A small number of outlier transactions extend up to around 350+ total amount, indicating occasional bulk or premium purchases.



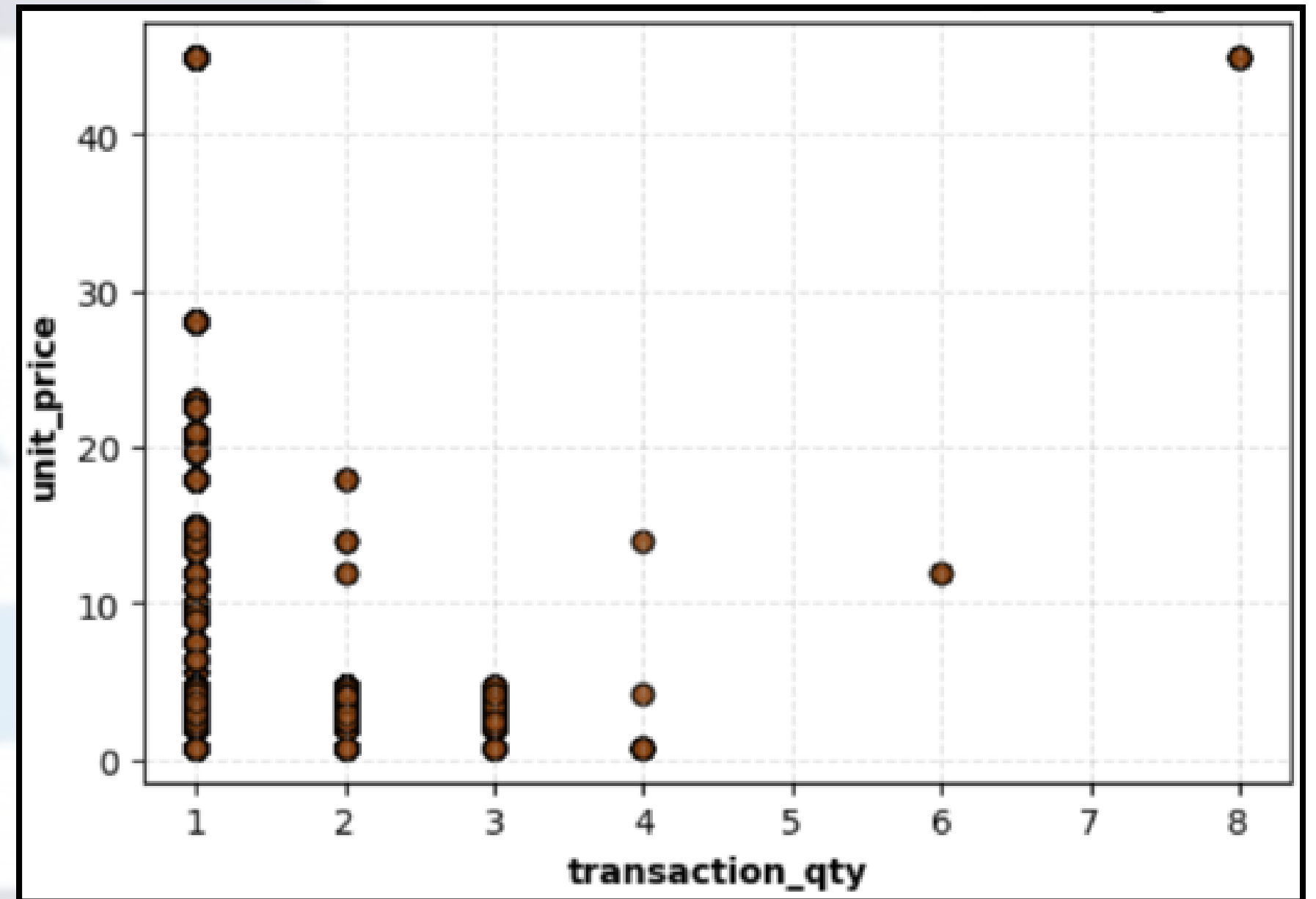
Distribution of Unit Price

- Most products have unit prices concentrated between the lower price ranges, indicating affordable product pricing.
- The distribution is right-skewed, meaning only a few products have very high prices.
- The majority of transactions occur for products priced around 2–5 units, showing these are the most commonly purchased items.
- A small number of products have prices extending up to 40+ units, indicating premium or specialty items.



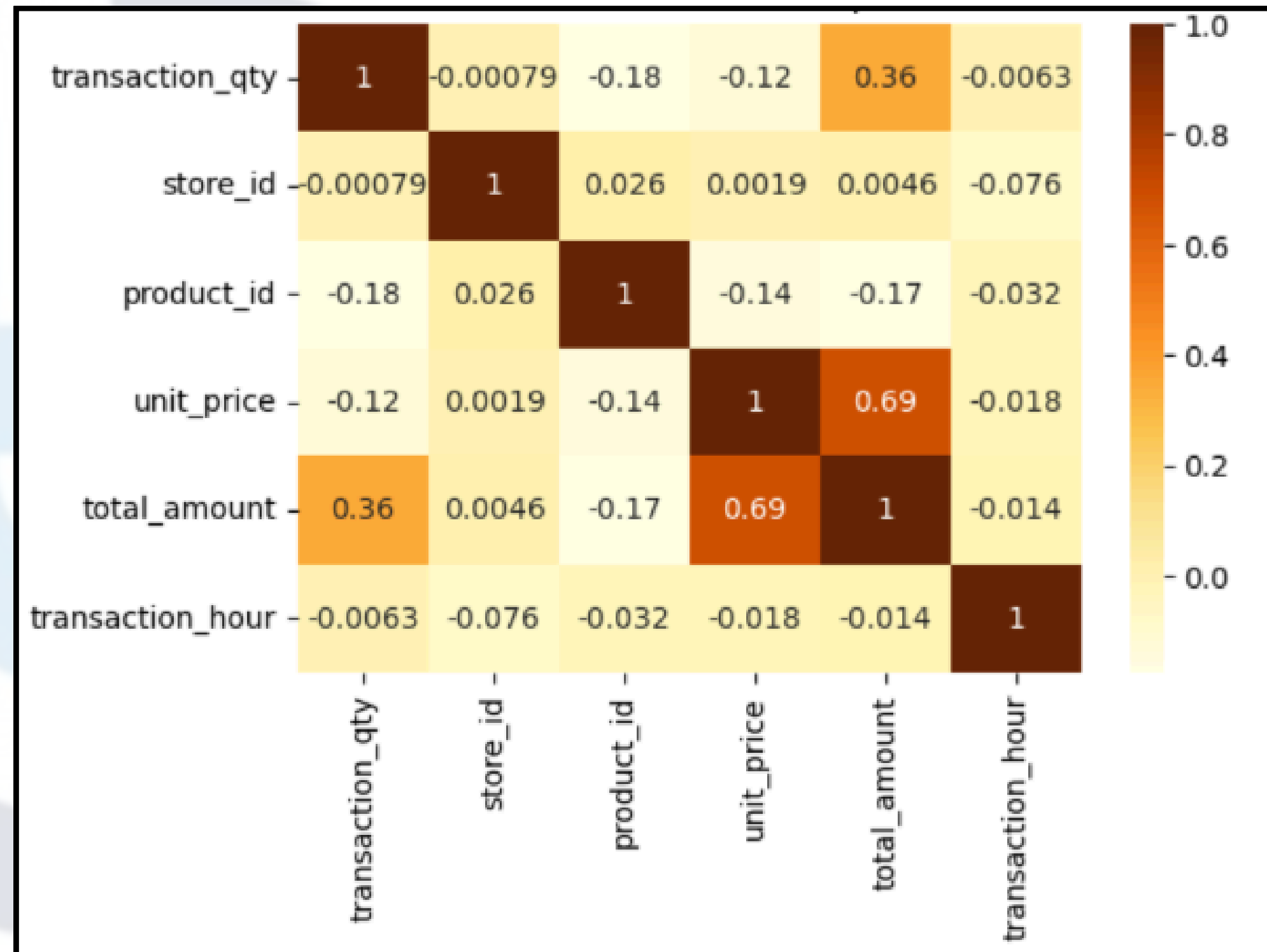
Unit Price vs Transaction Quantity

- Most transactions occur with lower quantities between **1 to 3 items**.
- Products with lower unit prices are purchased more frequently compared to expensive products.
- High-priced products (above 20 units) are generally bought in smaller quantities.
- A few outlier transactions show very high unit prices around 45 units, indicating premium products.
- There is no strong positive relationship between unit price and transaction quantity, suggesting customers usually buy limited quantities regardless of price.



Correlation Heatmap

- **Unit price has the strongest impact on total sales** with a high positive correlation of **0.69**.
- **Transaction quantity moderately increases total amount** with a correlation of **0.36**.
- **Store ID shows almost zero correlation** with other variables, meaning store differences do not strongly affect sales.
- **Transaction hour has very weak correlations**, so purchase timing does not greatly influence revenue or quantity.
- **Product ID has slight negative correlations** with quantity and total amount, indicating some products generate lower sales.
- **Overall insight:** Sales revenue is mainly influenced by product price and quantity sold.



A close-up, top-down view of a white ceramic cup filled with a latte. The coffee has a thick layer of foam with intricate latte art, including a heart shape and a leaf-like pattern. The cup is set on a dark, reflective surface, and the background is softly blurred.

Thank You